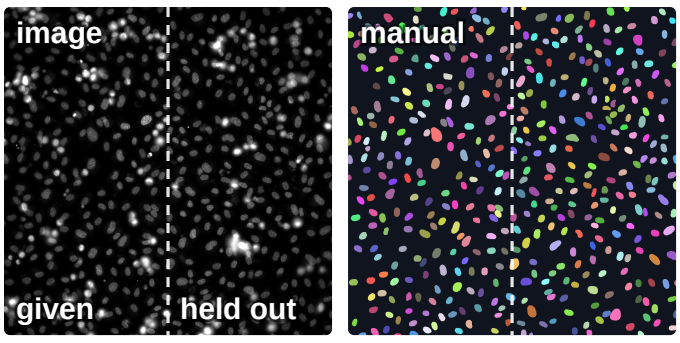
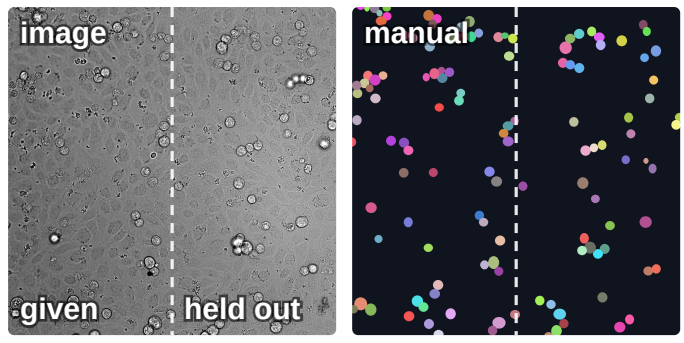
Nuclei



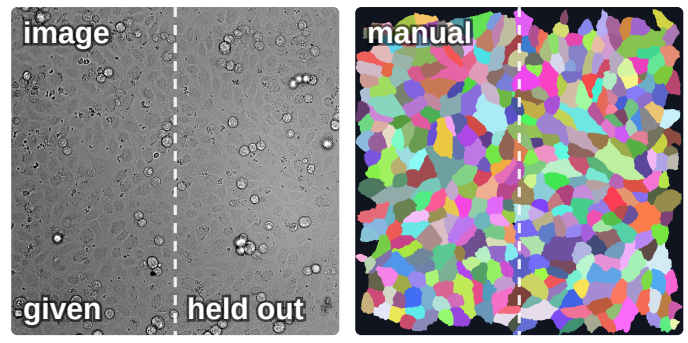
Segment and count the endothelial nuclei (not the cancer cells' nuclei)

Cancer cells



Segment and count the cancer cells

Endothelial cells



Segment and count the endothelial cells (whole cells, outlined at the junctions)

a Ask for advice

● outdated model or code suggested (one dot per reply, 3 per prompt and task)

basic prompt

"Which tool, model and settings would you recommend? I don't need code."

detailed prompt

+ "…check current documentation", model suitability, tests …

	Nuclei	Cancer cells	Endothelial cells
basic	● ● ● 3 of 3	○ ● ○ 1 of 3	○ ● ● 2 of 3
detailed	○ ○ ○ 0 of 3	○ ○ ○ 0 of 3	○ ○ ○ 0 of 3

"Cellpose (nuclei or cyto3 model)"
"The page I fetched doesn't mention the older nuclei model, so I can't confirm it still exists"

"Try Cellpose first (cyto3 or Cellpose-SAM)"
"Cellpose-SAM (Cellpose 4, cpsam) first"

"start with Cellpose 3 (cyto3) and fine-tune it"
"Cellpose 4 (Cellpose-SAM, cpsam)"

Outdated advice can fail or silently run another model. It checked the docs only when asked → ask it to check them

b Ask for code: a Python script

○ one run ✕ a field failed (counts 0) ■ median of the runs

basic prompt

"Please write a Python script to segment the …"

detailed prompt

Same request + explicit parameters, input checks and a test on reference data

	Nuclei	Cancer cells	Endothelial cells
basic	○ ○ 0.86	✕ ○ ○ 0.41	✕ ○ ○ 0.39
detailed	○ ○ ○ 0.76	✕ ○ ○ 0.51	✕ ○ ○ 0.69

Writing code doesn't guarantee it runs → ask for code that is easier to debug

Asking again gives a different script, but each script gives the same result every time it is run.

c Let an agent run it

○ one run ✕ a field failed (counts 0) ■ median of the runs

by eye

"…run the pipeline, assess, improve…"

scored

+ "…score every version against my annotation…"

	Nuclei	Cancer cells	Endothelial cells
by eye	○ ○ 0.96	○ ○ 0.90	✕ ○ ○ 0.48
scored	○ ○ 0.91	○ ○ 0.90	○ ○ 0.73

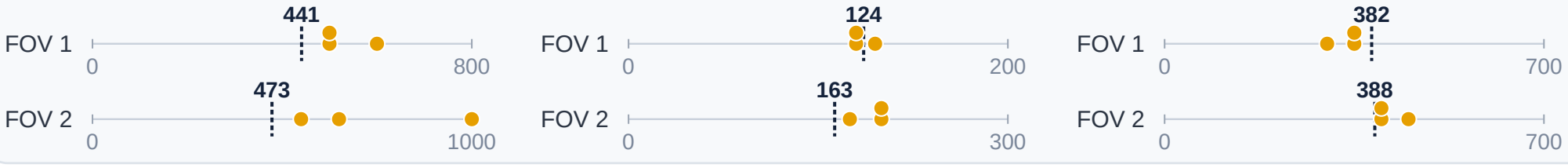
Without reference data the agent judges by eye → giving it a score to optimise may help on harder tasks

d Ask for the answer

● one run ⋮ manual count

"How many … are in this image?"

Asking again gives a different count: the answer is not reproducible



more of the analysis happens inside the LLM interaction

Regardless of the level of delegation, test on representative images with reference outputs

- **Check how the service handles data**
- **Verify advice against the docs**
- **Ask for code that is easy to debug**
- **Give agents a score to optimise**
- **Request code and intermediate outputs, not only the result**